Griffin Hill

Engineering Portfolio: https://griffin-hill.github.io/ | (936) 697-5225 | hillgri48@gmail.com

OBJECTIVE

Current Senior at the University of Oklahoma pursuing a Bachelor of Science in Computer Engineering, with Minors in Computer Science and Mathematics. Technical focus spanning embedded systems, hardware design, and software development. Experienced in embedded firmware development, PCB design, VLSI/CMOS logic and design, and analog/digital circuit design, with modern C++/Python development from a summer internship at Aspyr Media.

EDUCATION

Bachelor of Science in Computer Engineering, Minors in Computer Science and Mathematics
The University of Oklahoma | *Norman, OK* Expected Graduation: December 2026

Relevant Coursework
PCB Design, VLSI Systems Design, Operating Systems, Digital Design, Computer Architecture, Abstract/Numerical Linear Algebra, Machine Learning, Microprocessor System Design, Signals and Systems, Electronics Laboratory, Discrete Mathematics, Data Structures and Algorithms, and Circuits Laboratory.

WORK EXPERIENCE

Software Engineering Intern – Aspyr Media *Austin, TX | Summer 2026*

- Designed and built, from the ground up, a suite of Python libraries automating access to first-party development kits (Xbox, PlayStation, Nintendo) for automated game testing on the build server toolchain.
- Cut manual per-platform QA testing time from 45–90 minutes to 10–15 minutes by replacing manual tester workflows with automated builds.
- Worked independently to reverse-engineer integration requirements from limited, confidential first-party technical documentation, writing all code from scratch without generative-AI assistance.

Geek Squad Consultation Agent *Norman, OK | June 2025 – Present*

- Apple Authorized Service Provider: AST 2 diagnostics, RMA/warranty eligibility, privacy & consent.
- Communication: Technical translation for non-engineers, escalation, & documentation.

PROJECTS

- **Mesh Networking Wearable Device (Senior Capstone – In Progress):** Software developer on a 4-person team designing a self-contained, battery-powered wearable that logs event data locally and syncs it peer-to-peer across devices via BLE and LoRa radio, with no WiFi/cellular dependency, multi-day battery life, and no external antenna.
- **Bare-Metal STM32 Music Game:** Built a Simon Says–style rhythm game on an STM32G0 (ARM Cortex-M0+) in register-level C with no HAL, using SysTick-timed tone/chord synthesis, LFSR percussion noise, and UART score output.
- **Light Dimmer Controller:** Integrated 74HC595, 74HC165, EEPROM, and PWM for LED brightness control. Implemented persistent settings stored in EEPROM and custom 7-segment display encoding.

TECHNICAL SKILLS

- **Embedded Systems:** ARM Cortex-M, bare-metal microcontroller programming, register-level programming, timers & interrupts, I2C/SPI/UART, ADC/DAC interfacing.
- **Digital & Computer Architecture:** Processor data path design, pipelining, instruction set architecture, memory hierarchy, fixed/floating point arithmetic.
- **Circuit Design & Analysis:** Analog/digital circuits, circuit analysis, PCB design, schematics, and simulation software.
- **Machine Learning & Numerical Methods:** Numerical linear algebra, matrix factorizations (LU/QR/Cholesky), SVD, eigenvalue methods, ML model implementation & evaluation.
- **Software Development:** Modern C++, Python library/tooling design, build-server automation, cross-platform porting workflows (console/PC).
- **Tools:** Cadence Virtuoso/TSMC 16nm FinFET ADP, Altium Designer, KiCad, Multisim, ModelSim, STM32CubeIDE, DMM, logic analyzer, oscilloscope, Linux Terminal, MATLAB/Simulink, VS Code.
- **Programming:** C/C++, Java, Python, Verilog, Assembly, MATLAB, HTML/CSS, & DSA.

EXTRACURRICULAR ACTIVITIES

Phi Delta Theta, Oklahoma Alpha | VP, Treasurer, Alumni Chair *January 2024 – December 2025*

- Led a team of executive officers coordinating daily chapter operations, meetings, committees, and philanthropic initiatives as Vice President.
- Managed and budgeted an estimated $1.1 million in chapter finances as Treasurer, allocating funds across events, philanthropy, and chapter development.
- Organized alumni fundraising campaigns as Alumni Chair, securing support for facility improvements and scholarships.